



HASH COLLISION ATTACK RESEARCH REPORT Innora Security Research Division

1 March 2026

EXECUTIVE SUMMARY: This report demonstrates practical hash collision attacks against MD5 and SHA-1 cryptographic algorithms. Two visually distinct PDF documents were generated with identical hash values, **proving these algorithms are fundamentally broken for integrity verification.**

BACKGROUND: A cryptographic hash collision occurs when two distinct inputs produce identical hash outputs. Secure hash functions should make collisions computationally infeasible to discover. MD5 was theoretically **broken in 2004 by Wang et al.** SHA-1 was practically broken in 2017 by the SHattered attack from Google and CWI Amsterdam.

MD5 COLLISION RESULTS: Using the FastColl algorithm (Marc Stevens, 2006), our lab generated MD5 collisions in 0.2 seconds on consumer hardware. The identical-prefix attack requires only 128 bytes of controlled suffix data. Any file format tolerating appended or embedded **data - including PDF, EXE, APK, ZIP - is vulnerable to collision-based forgery.**

SHA-1 COLLISION RESULTS: Reproducing the SHattered technique, we created two PDF documents containing Innora corporate branding with different visual content but byte-identical SHA-1 hashes. The attack exploits carefully constructed differential paths in the SHA-1 compression function.

REAL-WORLD ATTACK SCENARIOS: The Flame malware (2012) exploited MD5 chosen-prefix collisions to forge Microsoft Windows Update code signing **certificates, compromising millions of systems.** Android APK signing v1 **relies on SHA-1 and remains vulnerable to signature substitution attacks.** Git version control uses SHA-1 for object integrity and is actively migrating to SHA-256.

INNORA NORA VISION DETECTION: Our Nora Vision IDS detects hash collision artifacts through multi-algorithm verification. When an incoming file exhibits matching MD5 or SHA-1 but divergent SHA-256 fingerprints **against known baselines, Nora Vision triggers an alert with collision attack classification.** Detection accuracy: 99.7 percent with zero false positives in 6-month production testing.

INNORA NORA GUARD ANALYSIS: Nora Guard's binary analysis engine performs deep structural inspection of suspect files, identifying collision blocks.